

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:50

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

(192425222)
I K. Sumanth Kumar Reddy (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

K. Sumanth
Signature of the student:

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APPLICATION- BASED QUESTIONS

1. Scenario: Event Communication Security: A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message "MEET AT NOON" is transformed into an encoded version using a fixed shift pattern.

- a. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
- b. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

2. Scenario: Intercepted Suspicious Message: A cybersecurity intern intercepts the coded message: "GSRH RH Z HVXIVG" during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

- a. Deduce the most likely original message.
- b. Justify your answer by analysing repeating patterns, letter structures, or known cipher characteristics.

3. Scenario: Secure Messaging System: A company deploys a keyword-based encryption technique for internal communication. An employee sends the word "HELLO" using the keyword "KEY" to ensure confidentiality.

- a. Demonstrate how the encryption process transforms the message step-by-step.
- b. Explain how the repeating keyword influences each character of the ciphertext.

4. Scenario: Suspicious Digital Asset Investigation: During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.

- a. Analyze how investigators could detect the presence of hidden data within the image.
- b. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

5. Scenario: Legacy Encryption in Corporate Communication: A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: "WECRLTEERDSOEFEAOCAIVDEN".

- a. Determine the most probable original message by reversing the encryption process.
- b. Critically evaluate why such a method is vulnerable in modern communication systems.
- c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25 20	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30 25	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20 20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15 15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10 10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

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1) Event Communication Security

The scenario describes a Caesar cipher, where every letter is shifted by a fixed number of positions in the alphabet.

a) Encryption using a shift of 4

Original message:

MEET AT NOON

Shift each letter 4 positions forward:

Original	shift +4	Encrypted
M	→ Q	Q
E	→ I	I
E	→ I	I
T	→ X	X
A	→ E	E
T	→ X	X
N	→ R	R
O	→ S	S
O	→ S	S
N	→ R	R

Therefore, the encrypted message is:

QIIX EX RSSR

b) Decryption by an authorized member

An authorized member who knows that the encryption uses a shift of 4 can decrypt the message by shifting every letter 4 positions backward.

Encrypted: QIIX EX RSSR

Shift - 4 : MEET AT NOON

For example:

- $Q \rightarrow M$
- $I \rightarrow E$
- $X \rightarrow T$
- $E \rightarrow A$
- $R \rightarrow N$
- $S \rightarrow O$

Thus, the original confidential message is:

"MEET AT NOON"

The main advantage of this method is its simplicity, but it provides very weak security because there are only a limited number of possible shifts.

Secure Messaging System

The company uses a keyword-based encryption technique, commonly represented by Vigenere cipher. In this method, the plaintext letters are combined with letters of a keyword. Here, the plaintext is "Hello" and the keyword is "key".

a) Encryption process step-by-step

Step 1: write the Plaintext

Plaintext: H E L L O

Step 2: Repeat the keyword until it matches the length of the message

Plain text: H E L L O

Keyword: K E Y K E

Step 3: Convert letters into numerical values

Position	Plaintext	Value	Keyword	Value
1	H	7	K	10
2	E	4	E	4
3	L	11	Y	24
4	L	11	K	10
5	O	14	E	4

Step 4: Add The Plaintext and Keyword values

The Vigenere encryption formula is:

$$C = (P + K) \bmod 26$$

where:

- C = cipher text value
- P = plain text value
- K = Keyword value

1. H+K:

$$7 + 10 = 17 \rightarrow R$$

2. E+E:

$$4 + 4 = 8 \rightarrow I$$

3. L+Y:

$$11 + 24 = 35$$

$$35 \bmod 26 = 9 \rightarrow J$$

4. L+K:

$$11 + 10 = 21 \rightarrow V$$

5. O+E:

$$14 + 4 = 18 \rightarrow S$$

Therefore,

Final encrypted message:

HELLO $\rightarrow$ RIJVS

b) How the repeating keyword influences each other

The keyword "Key" determines how much each plaintext character is shifted. Since the message is longer than keyword, the keyword is repeated cyclically.

The keyword therefore influences the encryption as follows:

- K shifts H by 10 positions $\rightarrow R$
- E shifts E by 4 positions $\rightarrow I$
- Y shifts L by 24 positions $\rightarrow J$
- The keyword starts again with K, so K shifts the second L $\rightarrow V$
- E then shifts O $\rightarrow S$

The important feature is that different keyword letters produce different shifts.

Thus, the keyword KEY creates pattern:

K $\rightarrow$ E $\rightarrow$ Y $\rightarrow$ K $\rightarrow$ E $\dots$

and transforms

HELLO $\rightarrow$ RIJVS

This makes the encryption more difficult to break.

For the message "Hello", the keyword "key" is repeated as follows:

Plaintext: H E L L O

Keyword: K E Y K E

↓ ↓ ↓ ↓ ↓

Ciphertext: R I J V S

Effect of keyword repetition

The repetition of KEY is important because it creates a changing encryption shift:

K → shift by 10

E → shift by 4

Y → shift by 24

K → shift by 10

E → shift by 4

Therefore, the same plaintext does not necessarily produce the same ciphertext letter. For example, if the letter L occurs at a position corresponding to Y, it becomes J, while another L corresponding to K becomes V.